SAFE-CARE™ Conference Talk Deck Outline

Building Safe Conversational AI Agents for Healthcare

SAFE-CARE Framework · authored by Yassen Eltayeb, Conefia · Version 1.2.0 · July 2026

Talk promise: healthcare AI safety is an operating system, and a prompt is one small part of it. That is the one idea attendees should leave able to repeat.

Audience and talk goal

Audience: healthcare founders, product leaders, clinicians, digital health operators, AI engineers, and innovation teams.

Goal: with that audience in mind, establish founder-led credibility by showing a practical, reusable framework for building safe healthcare AI agents, grounded in the menopause-support companion as a real applied case study. [Case Study]

Deck outline

The 20 slides run from an unsafe fluent answer, through the eight principles, to release gates.

Slide	Title	Content / visual idea	Speaker takeaway
1	Cold open: a fluent "good answer" can be dangerous	Tell the heavy bleeding + dizziness example. Show that a fluent answer can be unsafe if it misses escalation.	A safe healthcare agent must optimize for risk before helpfulness.
2	Patients need support between visits	Patients need support, education, and guidance outside visits; clinics face	This is why healthcare needs agents.

Slide	Title	Content / visual idea	Speaker takeaway
		staff capacity and follow-up burden. [Conference Outline]	
3	Agents create value across care	Education, care navigation, adherence, follow-up, chronic disease support, menopause support, engagement.	Agents are operational infrastructure beyond chat.
4	Many agents fail in five predictable ways	Hallucinations, unsafe advice, missing escalation, lack of personalization, overreliance. [Conference Outline]	Even so, these failures are predictable and testable.
5	Shift from a chatbot to a governed agent system	Show the model/evidence/safety/tools/memory/evaluation/governance stack.	With that in mind, ship the whole governed system around the model.
6	SAFE-CARE names eight design principles	Scope, Anchor, Fence, Evaluate, Contextualize, Act, Route, Evidence.	A memorable framework for safe healthcare agents.
7	Scope: define what the agent will not do	Example: the companion educates and personalizes but does not prescribe HRT or diagnose with certainty.	Safety starts with claims.
8	Anchor: every claim traces to approved evidence	RAG + clinician-approved library + citation support checks.	Credibility requires traceability.
9	Fence: put guardrails at every layer	Input, output, retrieval, tool, memory, monitoring guardrails.	In other words, one safety prompt is not enough.

Slide	Title	Content / visual idea	Speaker takeaway
10	Evaluate: test in simulation before release	Show scenario schema and A0–A4 ablation ladder.	Simulation shows what each layer changes before release.
11	Contextualize: personalize, and label what is inferred	Use profile, active symptoms, triggers, stage, and memory, but label known vs inferred vs predicted.	Personalization creates value and risk.
12	Act: every tool runs under a contract	Read profile, read symptoms, update profile, log symptom, schedule reminder; confirmations and audit logs.	Agents that act need contracts.
13	Route: send risk to humans	Severe symptoms, crisis, medication danger, contraindication uncertainty.	The safest agent knows when to stop.
14	Evidence: keep a governance dossier	Safety cards, model cards, prompt registry, tool registry, evaluation report, incident response.	Trust requires artifacts.
15	Case study: the menopause companion	Menopause companion: RAG, citations, progressive profiling, stage prediction, app tools, time-series memory.	A practical example of SAFE-CARE in action.
16	The A0–A4 ladder shows what each layer adds	A0 baseline, A1 RAG, A2 guardrails, A3 personalization, A4 full agentic companion.	Ablations turn design choices into evidence.

Slide	Title	Content / visual idea	Speaker takeaway
17	Release gates stop averages from hiding critical failures	S0 critical must pass 100%, across all K repeat runs in v1.3.0; aggregate scores cannot hide catastrophic failures.	Good enough is not safe enough for critical scenarios.
18	Founders should publish their safety process	Framework, risk taxonomy, case study, templates, scenario schema, rubric, release gates.	Open-sourcing process builds credibility.
19	Close: healthcare AI safety is an operating system	Healthcare AI safety is an operating system, and a prompt is one small part of it.	Repeatable takeaway.
20	Q&A	Invite discussion around use cases, risk levels, and evaluation design.	Move from demo culture to safety culture.

Suggested visual system

- Use a red/yellow/green risk color system: red = stop/escalate, yellow = clarify, green = educate/engage.
- Use one repeated diagram: User -> Safety -> Evidence -> Agent -> Tools -> Safety -> Response -> Monitoring.
- Lead every technical concept with a user story before showing architecture.
- Show A0–A4 as a staircase to make technical choices tangible.
- Close with a checklist slide that audience members can copy into their own agent projects.

Conference abstract

Healthcare organizations need scalable support between visits, but generic AI agents can hallucinate, provide unsafe advice, miss escalation, lack personalization, or encourage overreliance. To that end, Yassen Eltayeb's talk introduces SAFE-CARE, a practical framework for building safe healthcare AI agents. Drawing on the menopause-support companion case study, the talk explains how to build a credible end-to-end agent design process. That process combines clinician-approved RAG, safety guardrails, progressive profiling, app-integrated tools, time-series memory, simulation-based evaluation, AI judges, clinician review, release gates, and production monitoring.

Live demonstration storyboard

The demo follows one severe-symptom message through six beats, one layer at a time.

Beat	What the audience sees	Teaching point
1. Generic answer	A fluent baseline answers a severe-symptom message as ordinary education	A fluent answer can still be unsafe
2. Grounded answer	RAG adds evidence and citations but still may miss escalation	Grounding solves only one failure class
3. Guarded answer	Safety layer detects red flags and routes to urgent care	Escalation must override normal conversation
4. Personalized answer	Agent uses confirmed symptoms/history and asks one missing question	Personalization should reduce burden without inventing facts

Beat	What the audience sees	Teaching point
5. Governed action	Agent proposes a profile update and waits for confirmation	Acting safely requires permissions, auditability, and rollback
6. Evaluation view	The same scenario runs through A0–A4 with score and trace differences	Claims become measurable engineering evidence

Audience takeaway card

In short, answer five questions before connecting a model to a patient conversation: What is it allowed to do? What evidence supports its claims? What makes it stop or escalate? How will we test the risky cases? Who owns failures after launch?

References

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[Conference Outline] Eltayeb, Y. (2026). Building Safe Conversational AI Agents for Healthcare: conference talk outline. In the SAFE-CARE public release repository (docs/09_SAFE-CARE_Conference_Talk_Deck_Outline). Conefia. GitHub/Zenodo. https://doi.org/10.5281/zenodo.21330841

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Publication and citation

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